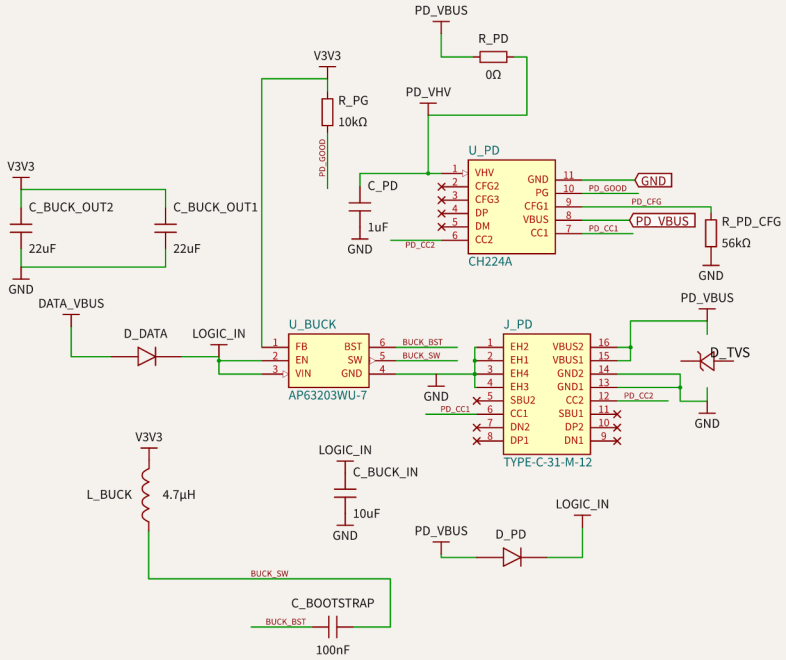
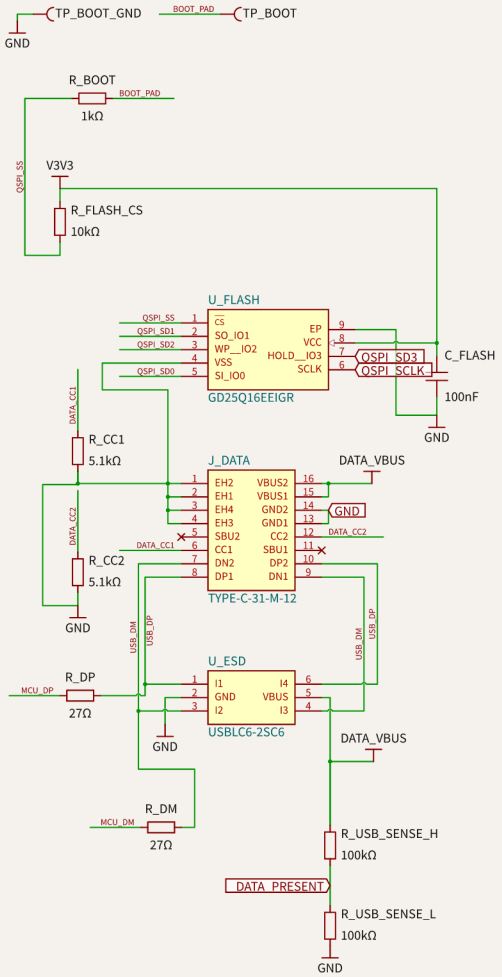
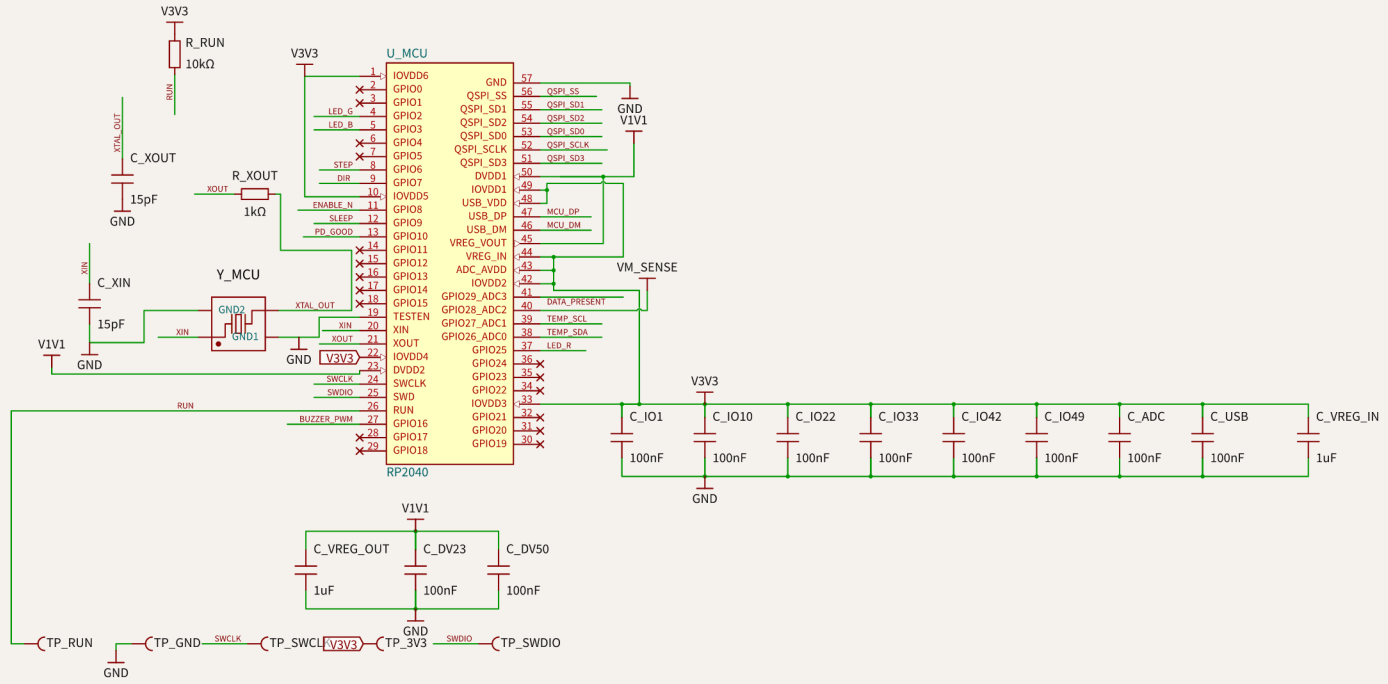
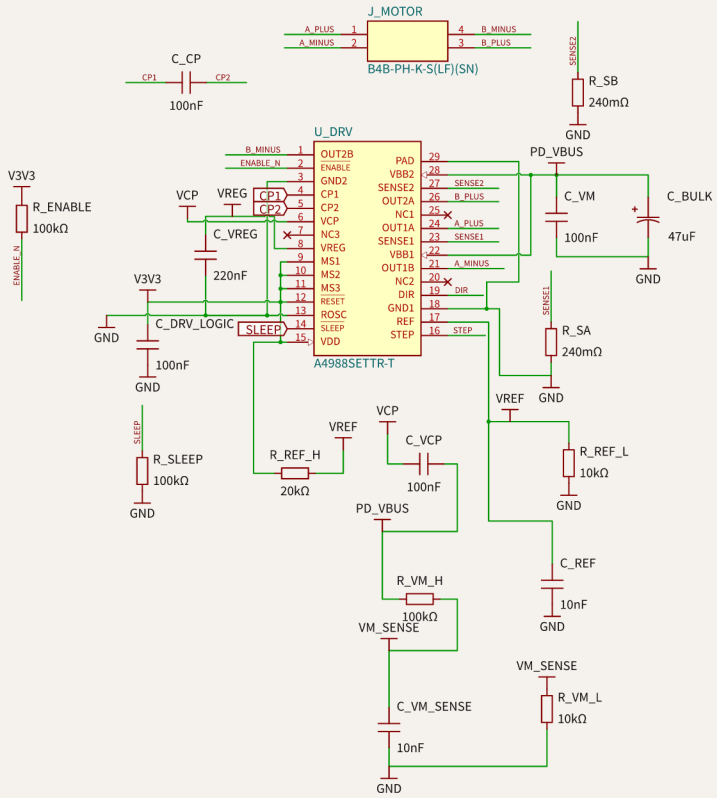
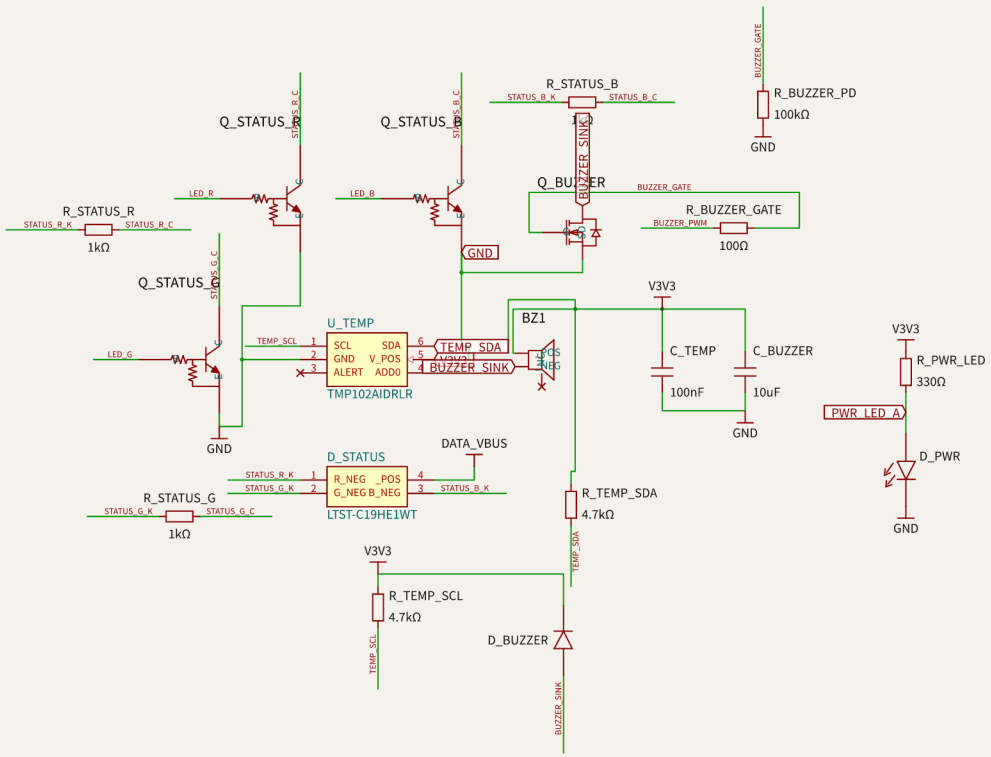
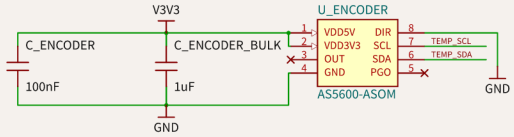

PD_VBUS
R_PD
0Ω
V3V3
R_PG
10kΩ
PD_VHV
PD_GOOD
U_PD
C_PD
1uF
GND
PD_CC2
VHV
CFG2
CFG3
DP
DM
CC2
GND
PG
CFG1
VBUS
CC1
PD_GOOD
PD_CFG
PD_CC1
CH224A
R_PD_CFG
56kΩ
V3V3
C_BUCK_OUT2
22uF
C_BUCK_OUT1
22uF
GND
DATA_VBUS
D_DATA
LOGIC_IN
U_BUCK
FB
EN
VIN
BST
SW
GND
BUCK_BST
BUCK_SW
AP63203WU-7
J_PD
EH2
EH1
EH4
EH3
SBU2
CC1
DN2
DP1
VBUS2
VBUS1
GND2
GND1
CC2
SBU1
DP2
DN1
PD_CC2
TYPE-C-31-M-12
PD_VBUS
D_TVS
GND
V3V3
L_BUCK
4.7µH
LOGIC_IN
C_BUCK_IN
10uF
GND
PD_VBUS
D_PD
LOGIC_IN
BUCK_SW
C_BOOTSTRAP
BUCK_BST
100nF


TP_BOOT_GND
BOOT_PAD
TP_BOOT
GND
R_BOOT
BOOT_PAD
1kΩ
QSPI_SS
V3V3
R_FLASH_CS
10kΩ
U_FLASH
QSPI_SS
QSPI_SD1
QSPI_SD2
QSPI_SD0
CS
SO_IO1
WP__IO2
VSS
SI_IO0
EP
VCC
HOLD__IO3
SCLK
QSPI_SD3
QSPI_SCLK
GD25Q16EEIGR
C_FLASH
100nF
GND
DATA_CC1
R_CC1
5.1kΩ
J_DATA
DATA_VBUS
EH2
EH1
EH4
EH3
SBU2
CC1
DN2
DP1
VBUS2
VBUS1
GND2
GND1
CC2
SBU1
DP2
DN1
GND
DATA_CC2
DATA_CC1
DATA_CC2
R_CC2
5.1kΩ
TYPE-C-31-M-12
USB_DM
USB_DP
USB_DM
USB_DP
U_ESD
R_DP
MCU_DP
27Ω
I1
GND
I2
I4
VBUS
I3
USBLC6-2SC6
GND
DATA_VBUS
R_DM
MCU_DM
27Ω
R_USB_SENSE_H
100kΩ
DATA_PRESENT
R_USB_SENSE_L
100kΩ
GND


V3V3
R_RUN
10kΩ
RUN
U_MCU
IOVDD6
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
IOVDD5
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
TESTEN
XIN
XOUT
IOVDD4
DVDD2
SWCLK
SWD
RUN
GPIO16
GPIO17
GPIO18
GND
QSPI_SS
QSPI_SD1
QSPI_SD2
QSPI_SD0
QSPI_SCLK
QSPI_SD3
DVDD1
IOVDD1
USB_VDD
USB_DP
USB_DM
VREG_VOUT
VREG_IN
ADC_AVDD
IOVDD2
GPIO29_ADC3
GPIO28_ADC2
GPIO27_ADC1
GPIO26_ADC0
GPIO25
GPIO24
GPIO23
GPIO22
IOVDD3
GPIO21
GPIO20
GPIO19
RP2040
LED_G
LED_B
STEP
DIR
ENABLE_N
SLEEP
PD_GOOD
XTAL_OUT
C_XOUT
15pF
GND
R_XOUT
1kΩ
XOUT
C_XIN
Y_MCU
GND2
GND1
V1V1
VM_SENSE
DATA_PRESENT
TEMP_SCL
TEMP_SDA
LED_R
C_IO1
C_IO10
C_IO22
C_IO33
C_IO42
C_IO49
C_ADC
C_USB
C_VREG_IN
100nF
1uF
C_VREG_OUT
C_DV23
C_DV50
TP_RUN
TP_GND
TP_SWCLK
TP_3V3
TP_SWDIO
SWCLK
SWDIO


J_MOTOR
A_PLUS
A_MINUS
B_MINUS
B_PLUS
B4B-PH-K-S(LF)(SN)
C_CP
CP1
CP2
100nF
R_SB
240mΩ
SENSE2
GND
U_DRV
B_MINUS
ENABLE_N
OUT2B
ENABLE
GND2
CP1
CP2
VCP
NC3
VREG
MS1
MS2
MS3
RESET
ROSC
SLEEP
VDD
PAD
VBB2
SENSE2
OUT2A
NC1
OUT1A
SENSE1
VBB1
OUT1B
NC2
DIR
GND1
REF
STEP
A4988SETTR-T
PD_VBUS
C_VM
100nF
C_BULK
47uF
V3V3
R_ENABLE
100kΩ
VCP
VREG
C_VREG
220nF
C_DRV_LOGIC
100nF
SLEEP
R_SLEEP
100kΩ
R_SA
240mΩ
SENSE1
VREF
R_REF_H
20kΩ
C_VCP
100nF
R_REF_L
10kΩ
C_REF
10nF
R_VM_H
100kΩ
VM_SENSE
C_VM_SENSE
10nF
R_VM_L
10kΩ


Q_STATUS_R
Q_STATUS_B
Q_STATUS_G
R_STATUS_B
R_BUZZER_PD
100kΩ
GND
BUZZER_GATE
BUZZER_SINK
Q_BUZZER
R_BUZZER_GATE
BUZZER_PWM
100Ω
R_STATUS_R
1kΩ
LED_R
LED_B
LED_G
U_TEMP
TEMP_SCL
TEMP_SDA
BUZZER_SINK
TMP102AIDRLR
BZ1
V3V3
C_TEMP
100nF
C_BUZZER
10uF
R_PWR_LED
330Ω
PWR_LED_A
D_PWR
D_STATUS
LTST-C19HE1WT
DATA_VBUS
R_STATUS_G
1kΩ
R_TEMP_SDA
4.7kΩ
R_TEMP_SCL
4.7kΩ
D_BUZZER


V3V3
C_ENCODER
100nF
C_ENCODER_BULK
1uF
GND
U_ENCODER
VDD5V
VDD3V3
OUT
GND
DIR
SCL
SDA
PGO
TEMP_SCL
TEMP_SDA
AS5600-ASOM
GND